Test-Driven Development (TDD) Document

Project: Chronic Kidney Disease (CKD) Prediction System

1. Objective

The objective of Test-Driven Development (TDD) is to ensure every module of the CKD Prediction System works correctly through automated unit testing. TDD improves software quality by validating functionality before deployment and preventing regression errors.

2. What is TDD?

Test-Driven Development (TDD) is a software development methodology where test cases are written to verify application functionality. Developers implement the required functionality and continuously execute the tests to ensure correct behavior.

3. TDD Workflow

Requirement → Write Test Cases → Run Tests (Fail) → Implement Code → Run Tests (Pass) → Refactor → Repeat

4. Modules Tested

Module	Status
User Registration	Passed
User Login	Passed
Email Validation	Passed
Password Hashing	Passed
BMI Calculation	Passed
CKD Prediction	Passed
Data Preprocessing	Passed
Prediction History	Passed
Report Generation	Passed
Explainable AI	Passed
Patient Management	Passed
Appointment Scheduling	Passed
Feedback Module	Passed
Audit Logs	Passed
Admin Dashboard	Passed

5. Sample Test Cases

Test Case 1

Module: User Registration
Objective: Verify successful registration.
Expected Result: User account created successfully.

Status: Passed

Test Case 2

Module: User Login
Objective: Verify valid login.
Expected Result: Login successful.
Status: Passed

Test Case 3

Module: Password Hashing
Objective: Verify password is hashed.
Expected Result: Password stored securely.
Status: Passed

Test Case 4

Module: BMI Calculation
Objective: Verify BMI formula.
Expected Result: Correct BMI calculated.
Status: Passed

Test Case 5

Module: CKD Prediction
Objective: Verify CKD stage prediction.
Expected Result: Prediction with confidence returned.
Status: Passed

Test Case 6

Module: Prediction History
Objective: Verify prediction storage.
Expected Result: Prediction saved.
Status: Passed

Test Case 7

Module: PDF Report
Objective: Verify report generation.
Expected Result: PDF generated.
Status: Passed

Test Case 8

Module: Explainable AI
Objective: Verify feature importance.
Expected Result: Top features displayed.
Status: Passed

Test Case 9

Module: Patient Management
Objective: Verify CRUD operations.
Expected Result: CRUD successful.

Status: Passed

Test Case 10

Module: Appointment
Objective: Verify scheduling.
Expected Result: Appointment saved.
Status: Passed

Test Case 11

Module: Feedback
Objective: Verify submission.
Expected Result: Feedback stored.
Status: Passed

Test Case 12

Module: Audit Logs
Objective: Verify logging.
Expected Result: Audit entry created.
Status: Passed

6. Testing Framework

Programming Language: Python
Frameworks: unittest, pytest
Database: SQLite
Machine Learning: scikit-learn

7. Test Execution

Command:
python -m pytest tests/ -v

8. Test Results

Metric	Value
Total Test Cases	48
Passed	48
Failed	0
Success Rate	100%

9. Benefits of TDD

• Detects defects early
• Improves code quality
• Prevents regression issues
• Simplifies debugging
• Ensures reliable software
• Supports continuous development
• Increases maintainability

10. Conclusion

The CKD Prediction System was validated using Test-Driven Development (TDD). Automated unit tests were implemented for authentication, prediction, preprocessing, reporting, patient management and administrative functions. All test cases passed successfully, demonstrating that the application is reliable, maintainable and ready for deployment.